CS4710: Artificial Intelligence
Homework 5

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Contents

1 Introduction **2**

2 Data processing and OLS model **2**
2.1 How did you preprocess the data (e.g., handling missing values, selecting variables)? 2
2.2 When fitting univariate linear regression models using different predictors, which predictor yields the best performance? 2
2.3 How does this best univariate model compare to the full multiple linear regression model? 3

3 Ridge regression **4**
3.1 Repeat the analysis in Question 1 using ridge regression. What is the optimal regularization parameter λ? 4
3.2 How does the model differ from or improve upon the OLS model? 4

4 Lasso regression **4**
4.1 Repeat the analysis in Question 1 using Lasso regression. What is the optimal regularization parameter λ? 4
4.2 How many coefficients are shrunk exactly to zero? 5
4.3 How does the model differ from or improve upon the OLS model? 5

5 Distribution analysis **5**
5.1 Are salary, height, and age normally distributed? 5
5.2 Does this result surprise you? Why or why not? 5

6 Extra credit **6**

7 Appendix: Python Output **7**

1 Introduction

This report analyzes the techSalaries2017 dataset using linear regression techniques, including ordinary least squares (OLS), ridge regression, and lasso regression. The goal is to understand the factors influencing total yearly compensation and evaluate model performance.

2 Data processing and OLS model

2.1 How did you preprocess the data (e.g., handling missing values, selecting variables)?

The dataset was first cleaned by replacing all "NA" entries with missing values and converting relevant variables to numeric format. To ensure proper interpretation of the education and race dummy variables, the dataset was restricted to rows where the qualitative variables "Education" and "Race" were not missing. Rows with remaining missing values in the predictors or the target variable were removed. Only quantitative predictors were used, and base salary, stock grants, and bonus were excluded to avoid trivial prediction of total compensation. One dummy variable was dropped from both the education and race groups to prevent multicollinearity. This approach ensures that the model uses valid and interpretable data, avoids misleading treatment of missing values, and prevents numerical instability. After preprocessing, the dataset was reduced from 62,642 rows to 21,681 usable observations, providing a consistent dataset for model training and evaluation.

2.2 When fitting univariate linear regression models using different predictors, which predictor yields the best performance?

Separate univariate linear regression models were trained using each predictor individually, and performance was evaluated using R^2 and RMSE on the test set. This was done to isolate the predictive power of each variable and determine which single factor best explains variation in compensation. The best-performing predictor, as shown in Figure 1, was `yearsofexperience`, achieving an R^2 of 0.1790 and RMSE of 118,624. Other strong predictors included SAT score ($R^2 = 0.1314$) and GPA ($R^2 = 0.1099$). This indicates that years of experience is the strongest individual predictor of compensation, although its explanatory power is still limited when used alone.

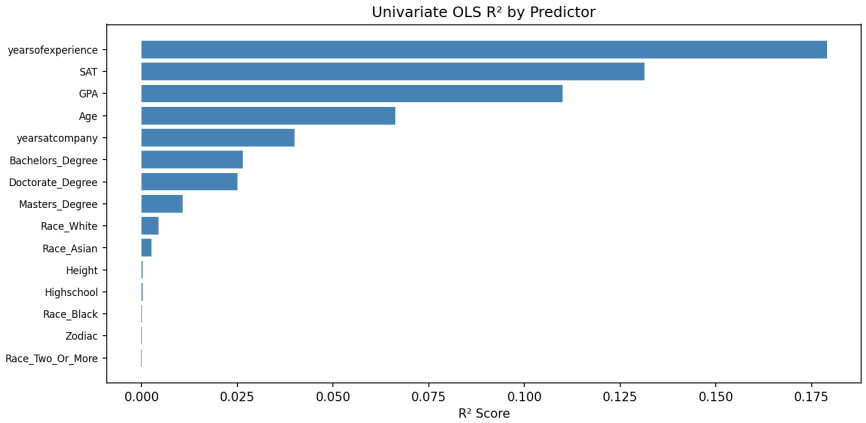

Figure 1: Univariate OLS performance across predictors

2.3 How does this best univariate model compare to the full multiple linear regression model?

A multiple linear regression model using all predictors was trained and evaluated on the test set to compare against the best univariate model. This allows us to assess the benefit of combining multiple variables instead of relying on a single predictor. As shown in Figure 2, the full OLS model achieved an R^2 of 0.2966 and RMSE of 109,803, which is a substantial improvement over the best univariate model. This demonstrates that compensation is influenced by multiple factors and that combining predictors significantly improves predictive performance.

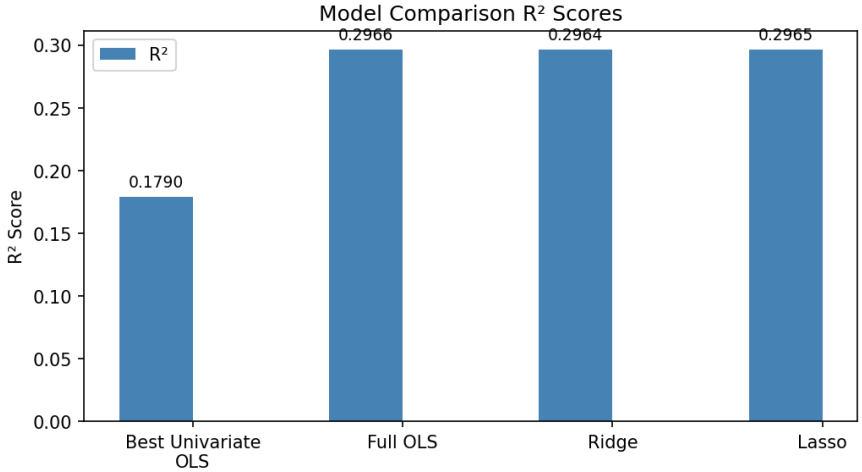

Figure 2: Comparison of model performance

3 Ridge regression

3.1 Repeat the analysis in Question 1 using ridge regression. What is the optimal regularization parameter λ?

Ridge regression was applied using standardized features, and the regularization parameter λ was tuned using cross-validation on the training data. This approach penalizes large coefficients and helps control overfitting. The optimal value was found to be $\lambda = 91.1163$. The ridge model achieved an R^2 of 0.2964 and RMSE of 109,815. This indicates that ridge regression performs similarly to OLS while introducing regularization to stabilize the model.

3.2 How does the model differ from or improve upon the OLS model?

The ridge regression model was compared directly with the OLS model to evaluate any improvements. Ridge regression produced nearly identical predictive performance to OLS, with only negligible differences in R^2 and RMSE. However, ridge shrinks coefficient magnitudes, reducing variance and making the model more stable. This suggests that while overfitting is not a major concern in this dataset, ridge regression still provides benefits in terms of coefficient regularization.

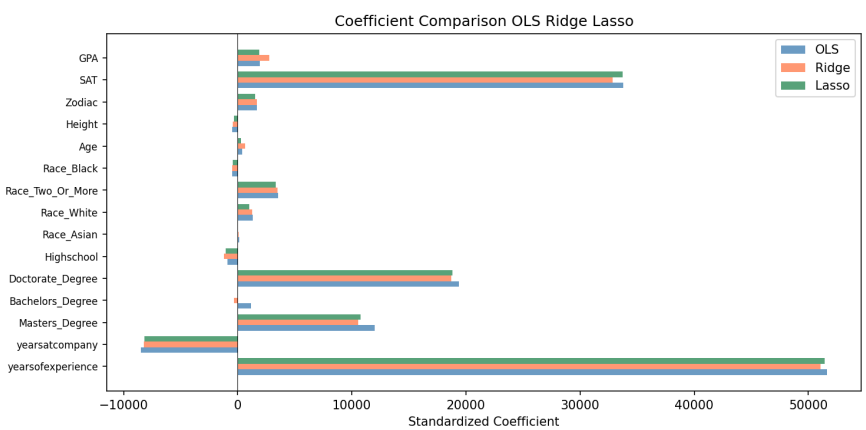

Figure 3: Coefficient comparison between OLS, Ridge, and Lasso

4 Lasso regression

4.1 Repeat the analysis in Question 1 using Lasso regression. What is the optimal regularization parameter λ?

Lasso regression was applied with standardized features, and the regularization parameter λ was tuned using cross-validation. Lasso introduces an L_1 penalty that can shrink coefficients to zero, effectively performing feature selection. The optimal value was found to be $\lambda = 123.2847$. The model achieved an R^2 of 0.2965 and RMSE of 109,809, similar to both

OLS and ridge regression. This shows that lasso maintains predictive performance while encouraging a simpler model.

4.2 How many coefficients are shrunk exactly to zero?

After training the lasso model, the number of coefficients equal to zero was counted. This was done to quantify the feature selection effect of lasso. The model shrunk 2 out of 15 coefficients exactly to zero. This indicates that these variables contribute little to predicting compensation and can be removed without loss of performance.

4.3 How does the model differ from or improve upon the OLS model?

The lasso model was compared with OLS to evaluate differences in performance and model structure. Lasso achieved nearly identical predictive performance to OLS but produced a sparser model by eliminating two predictors. This improves interpretability while maintaining accuracy, making lasso useful for simplifying the model.

5 Distribution analysis

5.1 Are salary, height, and age normally distributed?

The distributions of total yearly compensation, height, and age were analyzed using skewness, kurtosis (term learnt in another class), and statistical tests. Salary was found to be highly skewed (skewness = 4.51, kurtosis = 86.99) and clearly not normally distributed. Height showed near-zero skewness and kurtosis and appeared approximately normally distributed. Age had relatively low skewness (0.289) but failed formal normality tests, indicating it is not normally distributed. This analysis shows that only height closely follows a normal distribution.

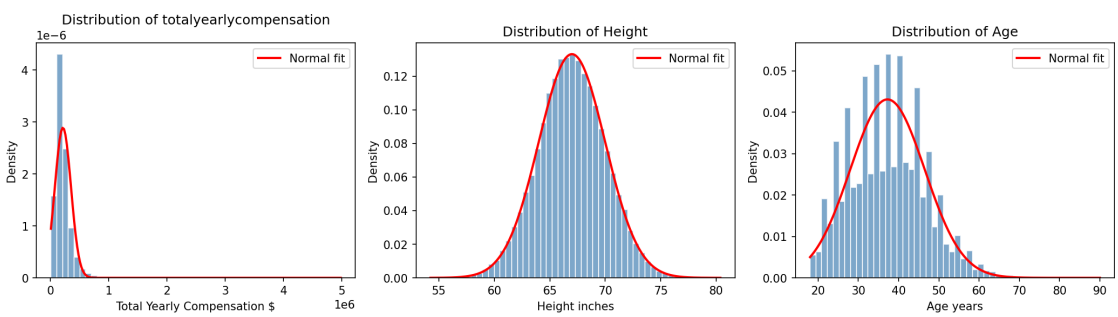

Figure 4: Distributions of salary, height, and age

5.2 Does this result surprise you? Why or why not?

These results are not surprising. Salary is expected to be right-skewed due to a small number of high earners creating a long tail. Height is a biological characteristic and typically

follows a normal distribution, which aligns with the observed results. Age reflects workforce demographics rather than a natural distribution, which explains why it deviates slightly from normality. Overall, the findings are consistent with real-world expectations.

6 Extra credit

An additional analysis was performed to examine the relationship between years of experience and compensation. A scatter plot with a fitted regression line shows a strong positive relationship, where compensation increases with experience but at a decreasing rate. This suggests diminishing returns to experience over time, which is a meaningful insight not directly captured by the regression models.

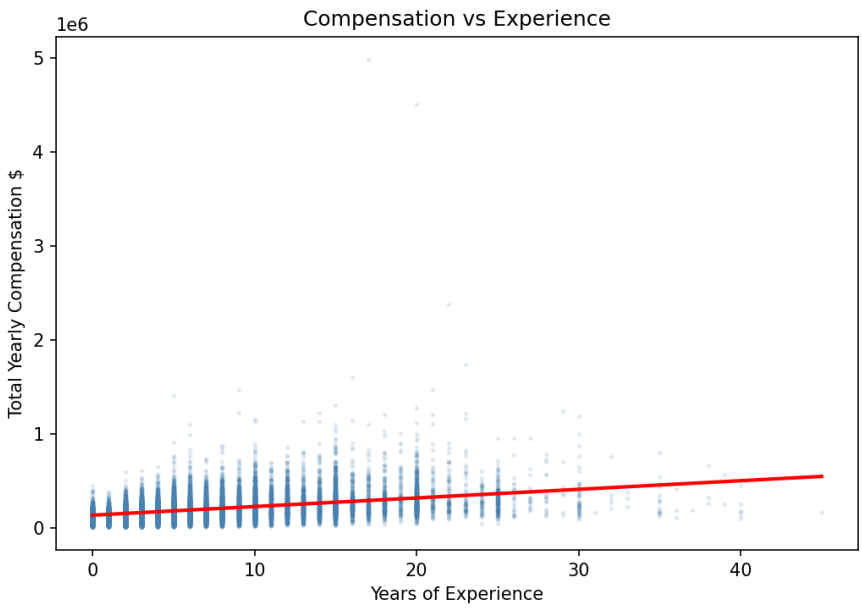

Figure 5: Compensation vs years of experience

7 Appendix: Python Output

The following is the console output obtained from running the `hw5.py` script, included to demonstrate the numerical results used in this report.

```
PS C:\Users\wangr\CS4710\Artificial-Intelligence> python .\hw5.py
Raw dataset: 62642 rows, 27 columns
After requiring Education and Race info: 21681 rows
After dropping remaining missing values in predictors and target: 21681 rows
Feature columns (15): ['yearsofexperience', 'yearsatcompany', 'Masters_Degree',
'Bachelors_Degree', 'Doctorate_Degree', 'Highschool', 'Race_Asian', 'Race_White',
'Race_Two_Or_More', 'Race_Black', 'Age', 'Height', 'Zodiac', 'SAT', 'GPA']
Train: 17344, Test: 4337

QUESTION 1 OLS REGRESSION

UNIVARIATE OLS MODELS
  yearsofexperience              R²=0.1790  RMSE=118,624
  yearsatcompany                 R²=0.0400  RMSE=128,276
  Masters_Degree                 R²=0.0107  RMSE=130,218
  Bachelors_Degree               R²=0.0264  RMSE=129,183
  Doctorate_Degree               R²=0.0250  RMSE=129,278
  Highschool                     R²=0.0002  RMSE=130,907
  Race_Asian                     R²=0.0026  RMSE=130,749
  Race_White                     R²=0.0044  RMSE=130,636
  Race_Two_Or_More               R²=-0.0002  RMSE=130,934
  Race_Black                     R²=0.0001  RMSE=130,912
  Age                            R²=0.0663  RMSE=126,505
  Height                         R²=0.0003  RMSE=130,903
  Zodiac                         R²=0.0001  RMSE=130,915
  SAT                            R²=0.1314  RMSE=122,019
  GPA                            R²=0.1099  RMSE=123,519

Best univariate predictor: yearsofexperience R²=0.1790 RMSE=118,624

Full OLS: R²=0.2966 RMSE=109,803

QUESTION 2 RIDGE REGRESSION
Optimal Ridge lambda: 91.1163
Ridge: R²=0.2964 RMSE=109,815

QUESTION 3 LASSO REGRESSION
Optimal Lasso lambda: 123.2847
Lasso: R²=0.2965 RMSE=109,809
Coefficients shrunk to zero: 2 out of 15
```

```
QUESTION 4 DISTRIBUTION ANALYSIS

totalyearlycompensation
  N=62642 Mean=216300.37 Std=138032.64
  Skewness=4.5125 Kurtosis=86.9951

Height
  N=62642 Mean=66.99 Std=3.00
  Skewness=-0.0013 Kurtosis=0.0012

Age
  N=62642 Mean=37.23 Std=9.26
  Skewness=0.2890 Kurtosis=-0.2111

SUMMARY
Model                            R²         RMSE
----------------------------------------------
Best Univariate OLS         0.1790      118,624
Full OLS                    0.2966      109,803
Ridge lambda=91.12          0.2964      109,815
Lasso lambda=123.28         0.2965      109,809

Lasso zeroed 2 of 15 coefficients
```